



RETAIL SALES ANALYSIS USING SQL

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PROJECT DESCRIPTION

This project analyses retail sales transaction data using SQL to extract meaningful business insights. The dataset includes transaction details such as product category, quantity sold, pricing, total sales amount, and customer demographics (age and gender). Using SQL queries, the data is analysed to evaluate overall sales performance, identify top-performing product categories, understand sales trends over time, and analyse purchasing behaviour across different demographic groups. The project demonstrates the use of SQL aggregation, grouping, and filtering techniques to support data-driven retail decision-making.

PROJECT OBJECTIVES

- To analyse overall retail sales performance using SQL.
- To identify top-performing product categories.
- To understand sales trends over time.
- To analyse customer purchasing behaviour based on age and gender.
- To generate meaningful insights for business decision-making.

DATASET OVERVIEW

The dataset used in this project contains retail sales transaction records. Each row represents a single transaction with details related to sales, product category, and customer demographics.

KEY COLUMNS:

- **Transaction_ID** – Unique identifier for each transaction
- **Date** – Date of purchase
- **Gender** – Gender of the customer
- **Age** – Age of the customer
- **Product_Category** – Category of the product purchased
- **Quantity** – Number of units sold
- **Price_Per_Unit** – Price of one unit
- **Total_Amount** – Total transaction value

TOOLS & SKILLS USED

TOOLS

- Microsoft SQL Server
- SQL Server Management Studio (SSMS)
- Microsoft PowerPoint

SQL SKILLS APPLIED

- Data Import (CSV to SQL Table)
 - Data Filtering using WHERE
- Data Aggregation using SUM, COUNT, AVG
 - Grouping Data using GROUP BY
- Sorting Results using ORDER BY
 - Date-based Analysis
- Data Cleaning and Validation

DATA IMPORT & INITIAL EXPLORATION

SQL QUERY:

- Created a dedicated database for retail sales analysis
- Loaded retail sales CSV data into SQL Server
- Performed initial data exploration using **SELECT *** to verify data integrity

Output:

- Successfully displayed transaction-level retail sales data
- Dataset contains customer demographics, product categories, pricing, and total sales values

```
create database retail_sales
use retail_sales
select * from [dbo].[retail_sales_dataset]
```

Results		Messages							
	Transaction_ID	Date	Customer_ID	Gender	Age	Product_Category	Quantity	Price_per_Unit	Total_Amount
1	1	2023-11-24	CUST001	Male	34	Beauty	3	50	150
2	2	2023-02-27	CUST002	Female	26	Clothing	2	500	1000
3	3	2023-01-13	CUST003	Male	50	Electronics	1	30	30
4	4	2023-05-21	CUST004	Male	37	Clothing	1	500	500
5	5	2023-05-06	CUST005	Male	30	Beauty	2	50	100
6	6	2023-04-25	CUST006	Female	45	Beauty	1	30	30
7	7	2023-03-13	CUST007	Male	46	Clothing	2	25	50
8	8	2023-02-22	CUST008	Male	30	Electronics	4	25	100
9	9	2023-12-13	CUST009	Male	63	Electronics	2	300	600
10	10	2023-10-07	CUST010	Female	52	Clothing	4	50	200
11	11	2023-02-14	CUST011	Male	23	Clothing	2	50	100
12	12	2023-10-30	CUST012	Male	35	Beauty	3	25	75
13	13	2023-08-05	CUST013	Male	22	Electronics	3	500	1500
14	14	2023-01-17	CUST014	Male	64	Clothing	4	30	120
15	15	2023-01-16	CUST015	Female	42	Electronics	4	500	2000
16	16	2023-02-17	CUST016	Male	19	Clothing	3	500	1500
17	17	2023-04-22	CUST017	Female	27	Clothing	4	25	100
18	18	2023-04-30	CUST018	Female	47	Electronics	2	25	50
19	19	2023-09-16	CUST019	Female	62	Clothing	2	25	50
20	20	2023-11-05	CUST020	Male	22	Clothing	3	300	900
21	21	2023-01-14	CUST021	Female	50	Beauty	1	500	500
22	22	2023-10-15	CUST022	Male	18	Clothing	2	50	100
23	23	2023-04-12	CUST023	Female	35	Clothing	4	30	120
24	24	2023-11-29	CUST024	Female	49	Clothing	1	300	300
25	25	2023-12-26	CUST025	Female	64	Beauty	1	50	50
26	26	2023-10-07	CUST026	Female	28	Electronics	2	500	1000
27	27	2023-08-03	CUST027	Female	38	Beauty	2	25	50
28	28	2023-04-23	CUST028	Female	43	Beauty	1	500	500
29	29	2023-08-18	CUST029	Female	42	Electronics	1	30	30
30	30	2023-10-29	CUST030	Female	39	Beauty	3	300	900

Query executed successfully.

TOTAL REVENUE ANALYSIS

SQL Query Description:

- Used SQL aggregate function SUM(total_amount) to compute overall revenue

```
--total revenue/sales--  
select sum(total_amount) as total_amount  
from [dbo].[retail_sales_dataset]
```

Output:

- Total Sales Revenue = 456,000

Key Insight:

- The dataset generated a total revenue of 456,000, representing the overall business sales performance
- This metric helps in evaluating the financial performance of the retail business

	total_amount
1	456000

SALES BY PRODUCT CATEGORY

SQL Query Description:

- Used SQL GROUP BY with SUM(total_amount) to calculate category-wise sales
- Sorted the results in descending order to identify top-performing categories

Output:

- Electronics: 156,905
- Clothing: 155,580
- Beauty: 143,515

Key Insight:

- Electronics is the highest revenue-generating category
- Clothing shows nearly similar performance, indicating strong demand
- Beauty contributes steadily to overall sales

```
--sales by product_category--  
select product_category,  
sum(total_amount) as total_category  
from [dbo].[retail_sales_dataset]  
group by Product_Category
```

	product_category	total_category
1	Clothing	155580
2	Electronics	156905
3	Beauty	143515

GENDER-WISE SALES ANALYSIS

SQL QUERY DESCRIPTION :

- Analysed total sales revenue based on customer gender
- Used SQL GROUP BY and SUM(total_amount) to calculate gender-wise sales

```
--sales by gender--  
select gender,  
sum(total_amount) as tota_sales  
from [dbo].[retail_sales_dataset]  
group by Gender  
order by Gender
```

OUTPUT:

- Female Customers: 232,840
- Male Customers: 223,160

KEY INSIGHT:

- Female customers contributed slightly higher sales revenue than male customers

	gender	tota_sales
1	Female	232840
2	Male	223160

AGE-WISE SALES ANALYSIS

SQL Query Description:

- Categorised customers into age ranges using CASE WHEN
- Aggregated sales using SUM(total_amount)

Output:

Sales data is grouped into age cate

- Under 25
- 25–34
- 34–55
- 45+

Key Insight:

- The 34–55 age group shows the highest sales contribution
- Customers aged 25–34 and 45+ also contribute significantly to overall revenue

```
-- sales by age analysis--
select
    case
        when Age < 25 then 'under_age'
        when Age between 25 and 34 then '25-34'
        when Age between 34 and 45 then '34-55'
        else '45+'
    end as age_group,
    sum(total_amount) as total_sales
from [dbo].[retail_sales_dataset]
group by Age
order by total_sales desc
```

	age_group	total_sales
1	34-55	17970
2	25-34	16785
3	45+	16065
4	under_age	14870
5	25-34	13980
6	under_age	13700
7	45+	13090
8	under_age	12585
9	45+	12505
10	34-55	11650
11	45+	11590
12	34-55	11290

✓ Query executed successfully.

Monthly Sales Analysis

SQL QUERY DESCRIPTION:

- Calculated total monthly sales using **SUM(total_amount)**
- Displayed the top 3 months based on sales data

OUTPUT:

- January (Month 1): 36,980
- February (Month 2): 44,060
- March (Month 3): 28,990

KEY INSIGHT:

- February recorded the highest sales among the analysed months
- Monthly analysis helps identify sales trends and seasonal patterns
- Useful for planning promotions and inventory management

```
--monthly sales--
select TOP 3
MONTH(Date) as month,
sum(total_amount) as total_monthly_sales
from [dbo].[retail_sales_dataset]
group by month(Date)
order by month
```

	month	total_monthly_sales
1	1	36980
2	2	44060
3	3	28990

Business Recommendations

- **Increase focus on Electronics and Clothing as top revenue categories**
- **Improve Beauty sales using discounts and combo offers**
- **Target female customers with personalised campaigns**
- **Focus marketing on the 34–55 age group, the highest contributor**
- **Plan promotions around high-performing months (February)**
- **Use regular SQL reports to track sales and customer trends**

PROJECT IMPACT

- **Provided clear insights into overall retail sales performance**
- **Identified top-performing product categories and customer segments**
- **Highlighted gender-wise and age-wise purchasing behaviour**
- **Revealed monthly sales trends to support better forecasting**
- **Demonstrated effective use of SQL for data-driven decision-making**

CONCLUSION

- **This project successfully analysed retail sales data using SQL**
- **Aggregate and grouping functions helped uncover key business insights**
- **Customer demographic analysis supported targeted marketing strategies**
- **Time-based analysis enabled understanding of sales patterns**
- **The project demonstrates strong foundational skills in SQL and data analysis**

THANK YOU